

SAVITRIBAI PHULE PUNE UNIVERSITY

Engineering Mathematics-I (BSC-101-BES)

2024 Credit Pattern | Semester I

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Q.1 is compulsory.
2. Attempt Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9, Q.10 or Q.11.
3. Neat diagrams must be drawn wherever necessary.
4. Assume suitable data, if necessary.
5. Use of electronic pocket calculator is allowed.

Q1) Write correct option for the following.

[10]

a) Stationary point for $f(x, y) = x^2 + y^2 + 6x + 12$ is: i) (1, 0) ii) (3, 0) iii) (-3, 0) iv) (0, 0) [2]

b) In the expansion of $\log(1 + e^x)$ in powers of x , coefficient of x is: i) 1 ii) 1/2 iii) 1/4 iv) 0 [2]

c) If

$$u = \frac{x^2 + y^2}{x + y}$$

, then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} =$$

: i) u ii) $2u$ iii) 0 iv) $3u$ [2]

d) Rank of matrix

$$\begin{bmatrix} 2 & -1 & -1 & 1 & 2 & 1 & 4 & -7 & -5 \end{bmatrix}$$

is: i) 3 ii) 2 iii) 1 iv) 0 [2]

e) If A is $n \times n$ with eigen values $\lambda_1, \lambda_2, \dots, \lambda_n$, then $\det(A) =$: i) [1]

$$\sum \lambda_i$$

ii)

$$\prod \lambda_i$$

iii) 0 iv)

$$n \cdot \prod \lambda_i$$

f) The quadratic form $3x^2 + 2y^2 + 3z^2 - 2xy - 2yz$ is: i) Positive definite ii) Negative definite [1]
iii) Indefinite iv) Semi-definite

Unit I --- Single Variable Calculus (14 Marks)

Q2) Attempt the following: [12]

a) Verify **Rolle's theorem** for $f(x) = (x - 1)(x - 2)(x - 3)$ in $[1, 3]$. [6]

b) Expand $e^x \sin x$ in powers of x using **Maclaurin's theorem** up to x^3 term. [6]

OR Q3) Attempt the following: [12]

a) Find the **Fourier series** of $f(x) = x$ in $(-\pi, \pi)$. Deduce that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \frac{\pi}{4}$$

. [6]

b) Find the half-range **sine series** for $f(x) = x(\pi - x)$ in $(0, \pi)$. [6]

Unit II --- Multivariable Calculus (14 Marks)

Q4) Attempt the following: [12]

a) If $u = x^2y + y^2z + z^2x$, show that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (x + y + z)^2$$

. [6]

b) If

$$u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$$

, prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

. [6]

OR Q5) Attempt the following: [12]

a) If $z = f(x, y)$ where $x = e^u \cos v$, $y = e^u \sin v$, prove that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{-2u} \left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \right)$$

. [6]

b) Find the **total derivative** of $u = \sin(xy)$ where $x = e^t$, $y = t^2$. [6]

Unit III --- Applications of Partial Differentiation (14 Marks)

Q6) Attempt the following: [12]

a) If $x = r \cos \theta$, $y = r \sin \theta$, find

$$\frac{\partial(x, y)}{\partial(r, \theta)}$$

and

$$\frac{\partial(r, \theta)}{\partial(x, y)}$$

. [6]

b) Find the **maximum and minimum** values of $f(x, y) = x^3 + 3xy^2 - 15x - 12y$. [6]

OR Q7) Attempt the following: [12]

a) Find the **stationary values** of $x^2 + y^2 + z^2$ subject to $x + y + z = 3a$ and $xy + yz + zx = 0$. [6]

- b) The period of a simple pendulum is $T = 2\pi\sqrt{l/g}$. If l is increased by 2% and g decreased by 3%, find percentage change in T . [6]

Unit IV --- Linear Algebra (14 Marks)

Q8) Attempt the following: [12]

- a) Find the **rank** of matrix

$$\begin{bmatrix} 1 & 2 & -1 & 3 & 2 & 4 & 1 & -2 & 3 & 6 & 3 & -7 \end{bmatrix}$$

.

[6]

- b) Solve the system: $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$ using **matrix inversion** method. [6]

OR Q9) Attempt the following: [12]

- a) Examine for **consistency**: $x + 2y + z = 2$, $2x - y - z = 2$, $4x - 7y - 5z = 2$. [6]
 b) Examine **linear dependence** of vectors: $(1, 2, -1, 3)$, $(2, 4, 1, -2)$, $(3, 6, 3, -7)$. [6]

Unit V --- Eigen Values (14 Marks)

Q10) Attempt the following: [12]

- a) Find **eigen values** and **eigen vectors** of

$$A = \begin{bmatrix} -2 & 2 & -3 & 2 & 1 & -6 & -1 & -2 & 0 \end{bmatrix}$$

.

[6]

- b) Reduce the **quadratic form** $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ to canonical form by **orthogonal transformation**. Find its nature. [6]

OR Q11) Attempt the following: [12]

- a) Verify **Cayley-Hamilton theorem** for

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \end{bmatrix}$$

. Hence find A^{-1} and A^3 .

[6]

- b) Diagonalize

$$A = \begin{bmatrix} 8 & -8 & -2 & 4 & -3 & -2 & 3 & -4 & 1 \end{bmatrix}$$

by finding modal matrix P .

[6]

*** END OF QUESTION PAPER ***